09-28 第2篇

阅读理解（说明文） Why Is Ice So Slippery?

① Every winter, millions of people around the world slip on ice. Children fall on the playground; old people break their wrists on the pavement; even professional ice-skaters sometimes spin across the rink almost without trying. But strangely, most of us cannot clearly answer a very simple question: **why exactly is ice slippery?** A solid stone is not slippery; a solid block of wood is not slippery. Why should a solid block of frozen water be so different? For more than one hundred and fifty years, this question has interested famous scientists, and the answer turns out to be **more complicated and more beautiful** than one might expect.

② For a long time, the most common explanation was **"pressure melting"**. In the 1850s, the British scientist **Lord Kelvin** and his brother suggested that the weight of a skater's thin blade pressed so hard on the ice that it briefly **melted a thin layer of water** under the blade. That thin layer of water, they said, was what made skating possible: the skater was actually **gliding on water**, not on ice. For nearly a century, this idea was taught in physics textbooks around the world.

③ However, by the 1930s, careful measurements showed that pressure alone **cannot** explain slippery ice. A heavy skater on a thin blade does press hard, but the **temperature drop** caused by that pressure is too small — it can melt ice only at temperatures just below 0 °C. Yet ice remains very slippery even at **−20 °C** or **−30 °C**, where Kelvin's pressure theory clearly fails. So scientists knew that another explanation was needed.

④ A second idea, proposed in the 1930s by British scientist **F. P. Bowden**, is called **"friction melting"**. When a skate blade moves across ice, the **rubbing** between the metal and the ice produces a small amount of heat. That heat melts a very thin layer of water, the blade slides on this water, and the ice feels slippery. This theory works well for moving objects, but it does not quite explain why **standing still on ice** is also slippery — after all, there is no rubbing when you are not moving, yet the ice is still dangerous under your feet.

⑤ The modern answer, supported by experiments since the 1990s, points to something **truly surprising**: even when ice is perfectly still, perfectly cold, and no-one is walking on it, its **top layer** is not quite solid. Very thin and delicate, this layer is often called a **"quasi-liquid layer"** (a "nearly-liquid" layer). In it, the water molecules are loosely held and can slide past each other — much like water can. This thin liquid-like film is **only a few molecules thick** at very low temperatures, but it is enough to make the ice act like a floor lightly covered with oil.

⑥ Why does ice naturally form this strange quasi-liquid layer, while a solid stone does not? The answer lies in the **shape of water molecules**. Water molecules in ice form a neat, open framework, like a delicate fence of small houses. But at the very top of the ice, there are **no more molecules above** to hold the surface molecules in place, so they are not locked as tightly as those in the middle. Even below 0 °C, these top molecules keep **wiggling**. The lower the temperature, the thinner this quasi-liquid layer becomes; but it never fully disappears until temperatures fall close to **−250 °C**, colder than any place on Earth.

⑦ The three explanations — pressure melting, friction melting, and the quasi-liquid layer — are **not really enemies**. Today, scientists believe all three play a role, but in different conditions. The quasi-liquid layer explains why ice is **always** slippery. Friction melting adds extra water when you **move** quickly across it. Pressure melting may play a **small** role only when very sharp blades cut into the ice at temperatures just below 0 °C.

⑧ From this story we can take two small lessons. First, **a simple question is not always simple**. "Why is ice slippery?" sounds like a childish question, but answering it fully takes solid state physics, chemistry, and many years of patient experiments. Second, **old answers are not always wrong**; they are sometimes only incomplete. Lord Kelvin's idea was not a mistake — it was a first step towards a larger truth. Science, at its best, is not about tearing down earlier thinkers, but about **standing on their shoulders and seeing a little further**.

注释： skate blade 冰刀；pressure melting 压融；friction 摩擦；quasi-liquid 类液态的；molecule 分子

根据文章内容，选择最佳答案。

1. What was Lord Kelvin's "pressure melting" idea, and why was it later challenged (Paragraphs 2—3)?

A. He said pressure from a skate blade melts a thin water layer that the skater glides on, but later measurements showed that pressure alone cannot explain why ice remains slippery at very low temperatures like −20 °C or −30 °C.
B. He said that cold air creates water on ice.
C. He said that skaters must be very strong to melt ice.
D. He said that ice is never slippery.

2. What does "friction melting" (Bowden's idea) propose (Paragraph 4)?

A. Friction from a moving blade against ice creates heat, which melts a thin layer of water, so the blade glides; but this does not fully explain why standing still on ice is also slippery.
B. Friction creates tiny sparks that melt the ice.

C. Skaters create friction by singing loudly.
D. Friction is only important at very high temperatures.

3. What does the modern "quasi-liquid layer" explanation say (Paragraphs 5—6)?

A. Ice is only solid when someone looks at it.
B. Even when ice is perfectly still, its top layer is a very thin "nearly-liquid" film, because surface water molecules have no molecules above them to lock them in place; this thin liquid-like film makes ice slippery.
C. The quasi-liquid layer only appears in Antarctica.
D. The quasi-liquid layer is made of oil.

4. How do the three explanations fit together in modern science (Paragraph 7)?

A. Only pressure melting is correct; the other two are wrong.
B. They are not enemies: the quasi-liquid layer explains why ice is always slippery; friction melting adds extra water when one moves fast; pressure melting may play a small role when sharp blades cut ice just below 0 °C.
C. All three are wrong and the truth is unknown.
D. Kelvin's theory is the only one that is still believed.

5. What is ONE lesson the writer draws at the end (Paragraph 8)?

A. Simple questions can require complex science, and old answers are not always wrong but sometimes only incomplete — science is about standing on earlier thinkers' shoulders and seeing a little further.
B. Children should stop asking questions about ice.
C. Skating should be banned in winter.
D. Every old theory should be thrown away.

答案与解析

1.
【答案】A
【题型】细节理解题
【定位】第②③段 "Lord Kelvin ... melted a thin layer of water ... too small ... fails at −20 °C"
【解析】Kelvin 压融论后被低温数据否定。B、C、D 均错。故选 A。

2.
【答案】A
【题型】细节理解题
【定位】第④段 "rubbing ... produces a small amount of heat ... does not quite explain why standing still on ice is also slippery"
【解析】Bowden 摩擦生热能解释滑动但不能解释静止。B、C、D 均错。故选 A。

3.
【答案】B
【题型】细节理解题
【定位】第⑤⑥段 "quasi-liquid layer ... no more molecules above ... keep wiggling"
【解析】表层分子无上方束缚呈准液态。A、C、D 均错。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑦段 "not really enemies ... all three play a role"
【解析】三种解释互补并非对立。A、C、D 均错。故选 B。

5.
【答案】A
【题型】主旨/观点题
【定位】第⑧段 "A simple question is not always simple ... standing on their shoulders and seeing a little further"
【解析】末段总结科学之进化。B、C、D 均错。故选 A。